FOUR EQUIVALENT CHARACTERIZATIONS OF ALGEBRAICALLY CLOSED FIELDS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The chapter's own definition-algebraically-closed (source lines 1117-1121, triviality of every algebraic extension) is characterization (1); the author notes before the lemma that other texts use the root formulation instead.

Editorial note. The label lemma-algebraically-closed names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter: example-degree-simple-algebraic-extension (a root of an irreducible polynomial of degree d gives the extension k[x]/(P) of degree d, so an irreducible of degree > 1 yields a nontrivial finite algebraic extension); division with remainder in F[x]

Source reference. Original reference 09G7: sources/stacks-fields/pinned-original-source.tex, source lines 721–721. Prerequisite pointer invoked at 1142; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let F be a field. The following are equivalent*

(1) *F is algebraically closed,*
(2) *every irreducible polynomial over F is linear,*
(3) *every nonconstant polynomial over F has a root,*
(4) *every nonconstant polynomial over F is a product of linear factors.*

Proof. If F is algebraically closed, then every irreducible polynomial is linear. Namely, if there exists an irreducible polynomial of degree > 1, then this generates a nontrivial finite (hence algebraic) field extension, see Example 09G7. Thus (1) implies (2). If every irreducible polynomial is linear, then every irreducible polynomial has a root, whence every nonconstant polynomial has a root. Thus (2) implies (3).

Assume every nonconstant polynomial has a root. Let $P \in F[x]$ be nonconstant. If $P(\alpha) = 0$ with $\alpha \in F$, then we see that $P = (x - \alpha)Q$ for some $Q \in F[x]$ (by division with remainder). Thus we can argue by induction on the degree that any nonconstant polynomial can be written as a product $c \prod (x - \alpha_i)$.

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Finally, suppose that every nonconstant polynomial over F is a product of linear factors. Let E/F be an algebraic extension. Then all the simple subextensions $F(\alpha)/F$ of E are necessarily trivial (because the only irreducible polynomials are linear by assumption). Thus $E = F$. We see that (4) implies (1) and we are done. $\square$